MIL-C-52093C(ME)
17 May 1983

SUPERSEDING
MIL-C-52093B(ME)
22 October 1969

MILITARY SPECIFICATION

CRADLE: PONTON BOAT, HALF-SECTION, ALUMINUM HULL

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cradle used to transport and to stow aluminum half-section ponton boats.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-461	- Wire, Steel, Carbon, (Round, Bare, and Coated).
TT-V-119	- Varnish, Spar, Phenolic-Resin.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materials.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein.

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DRAWING

ME

TA13217E1990 - Cradle, Ponton Boat Assembly.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The ponton boat cradle shall be as shown on Top Assembly TA13217E1990 and as specified herein.

3.1.1 Drawings. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawing to insure correct fit, assembly, and operation of the cradle. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subject to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The cradle shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Identification marking. The cradle shall be identified in accordance with MIL-STD-130 and as shown on the drawing.

3.5 Cleaning. All ferrous metal surfaces requiring painting shall be cleaned to the base metal by sandblasting in accordance with MIL-T-704, to remove all oil, grease, mill scale, products of erosion, dirt, welding slag, or spatters, and other foreign substances that may be detrimental to the coating system.

3.6 Workmanship. All parts, components, and assemblies of the cradle, including welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges and corners are required.

3.6.1 Steel fabrication. Steel used in the fabrication of the equipment shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or jigs to insure uniformity of size and shape.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured and examined in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article cradle. The first article cradle shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each cradle shall be examined for defects as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cradle shall be examined as specified herein for the following defects:

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Major

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Cradle not true to dimensions, contains twists, bends, or other imperfections.
106. Assembly defects, nonuniformity of parts and sections, ponton bearings not in same plane.
107. Welding and weldments not in accordance with the requirements on the drawing.
108. Cleaning and painting not as specified.
109. Workmanship not as specified.
110. Identification marking missing, incomplete, or illegible.

4.6 Inspection of packaging. Each cradle shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

No.	Defect	A	Comm.
111.	Cradle not preserved as required.	5.1.1	5.1.2
112.	Cradle not packed as required.	5.2.1	5.2.2
113.	Cradle not marked as required.	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative applicability. The provisions of MIL-P-116 shall apply when determining the applicability of preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

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5.1.1.2 Exterior surfaces. Unpainted exterior ferrous metal surfaces of bolts, nuts, pins, washers, chains, and any other exterior surface requiring a contact preservative (see 5.1.1.1), shall be coated with type P-1 preservative. Unpainted ferrous metal information plates shall be coated with varnish conforming to TT-V-119 or MIL-V-13811.

5.1.2 Commercial. The cradle shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each cradle shall be prepared for shipment loose (unboxed). The spreader rod and ramp assembly shall be secured to the cradle with the pins provided. The chains and binders shall be secured to the cradle with soft annealed wire conforming to QQ-W-461, finish 5, class 2, composition of either AISI 1006, 1008 or 1010. The diameter of the wire shall be not less than 0.0625 inches.

5.2.2 Commercial. The cradle shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Level A. In addition to any special or other identification marking required by the contract or purchase order (see 6.2), each cradle shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Each cradle shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The cradle is intended for either transporting ponton boats of the light tactical floating bridges raft section on 2-1/2 ton pole trailers or for storing them.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

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c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

d. When additional marking is required (see 5.3.1).

6.3 First article. When a first article inspection is required, the item shall be examined and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
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Preparing activity:
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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52093C(ME)	Cradle: Ponton Boat, Half-Section, Aluminum Hull

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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